HW6 - Explainable AI

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1 LIME on `toydataset.csv`

The dataset has 5000 observations with two continuous features $x_1, x_2 \in [-1, 0.5]$, one categorical feature $x_3 \in \{1, 2, 3\}$ (counts 1653/1739/1608), and a continuous target $y \in [0.10, 3.33]$ with $\bar{y} = 1.52$. The marginal $\bar{y} \mid x_3$ already separates the categories (1.84/1.36/1.35 for $x_3 = 1/2/3$), and the colour bands in Fig. 1 show that y depends non-monotonically on x_1 (high y near both $x_1 \approx 0.5$ and $x_1 \approx -1$) and is only weakly dependent on x_2.

The black-box is a Random Forest regressor (500 trees, `min_samples_leaf`=5, seed 42) fit on the raw (x_1, x_2, x_3) matrix. RF needs no scaling and handles ordinal-encoded x_3 by splitting on it directly, so the same representation is used by both the model and the LIME sampler. With an 80/20 train/test split, 5-fold CV on the train side gives $R^2 = 0.866 \pm 0.013$ (MSE 0.046 ± 0.003), and the held-out set gives $R^2 = 0.878$ (MSE 0.042). The model is the target of explanation; no claim is made that it recovers the data-generating process.

For each query point x, LIME draws $N = 5000$ perturbed points. Continuous features are sampled independently from $\mathcal{N}(\mu_j, \sigma_j^2)$ using training statistics, and x_3 is sampled from its empirical frequencies. This is the global sampling scheme of the reference LIME library; an alternative is $\mathcal{N}(x_j, \sigma_{\text{loc}}^2)$ centred on the instance, which gives higher fidelity but more variance across seeds. The interpretable representation is $z \in \{0, 1\}^3$: continuous features are discretized into training-set quartiles with $z_j = 1$ iff the perturbed value falls in the same bin as x, and for x_3, $z_j = 1$ iff the perturbed category equals x_3. Coefficients then read directly as "being in the instance's bin for feature j contributes this many y-units". Four bins balance resolution against samples per cell; finer binning would shrink each cell's effective sample size.

Sample weights are $\pi_x(x') = \exp(-d^2/\sigma_K^2)$ with d Euclidean in standardized space (continuous features in z-score units, categorical mismatch contributing one unit of distance) and $\sigma_K = 0.75\sqrt{d_{\text{feat}}}$, the LIME default, which sets the effective neighbourhood radius. The surrogate is a weighted Lasso with $\alpha = 0.01$, small enough to retain non-trivial effects but large enough to zero out features with no local signal. Repeating the explanations under 5 different seeds gives coefficient std ≤ 0.025 on every instance, so what follows is well-determined.

Table 1: LIME explanations of the four query points. Coefficients β_j are those of the weighted Lasso on the binary z. The instance has $z = (1, 1, 1)$ by construction, so $g(x) = \beta_0 + \sum_j \beta_j$.

x_1	x_2	x_3	$f(x)$	β_0	$\beta_{x_1 \text{ bin}}$	$\beta_{x_2 \text{ bin}}$	$\beta_{x_3=\text{cat}}$
0.4	-0.4	1	2.20	1.02	+0.50 (top)	0.00 (Q2)	+0.41 (x_3=1)
0.2	-0.4	1	1.36	1.09	+0.30 (top)	0.00 (Q2)	+0.42 (x_3=1)
0.4	-0.4	2	1.65	1.21	+0.48 (top)	0.00 (Q2)	-0.17 (x_3=2)
0.4	0.2	2	1.66	1.21	+0.44 (top)	0.00 (top)	-0.19 (x_3=2)

All four instances have $x_1 > 0.13$, the top quartile, and LIME assigns this bin a positive coefficient (+0.30 to +0.50), matching the high-y band at $x_1 \approx 0.5$ visible in Fig. 1. The x_2 coefficient is shrunk to zero by Lasso on every instance: there is no local x_2 gradient at these query points, consistent with the weak visual x_2 dependence. The x_3 coefficient is +0.4 when the instance has $x_3 = 1$ and -0.18 when $x_3 = 2$, matching the marginal $\bar{y} \mid x_3$ gap.

One limitation of quartile-LIME is visible in the table. The first two instances share an identical $z = (1, 1, 1)$ because both have x_1 in the top quartile, x_2 in Q2, and $x_3 = 1$, so the surrogate assigns them the same explanation, yet $f(x)$ differs by 0.84. This is the source of the low local R^2 (0.17 to 0.29 across instances): the binary surrogate cannot resolve the $x_1 = 0.4 \to 0.2$ gradient inside the top quartile, where the data-overview plot shows a steep drop. Refusing discretization, and using z-scored x' directly as surrogate features instead, trades this readability for fidelity.

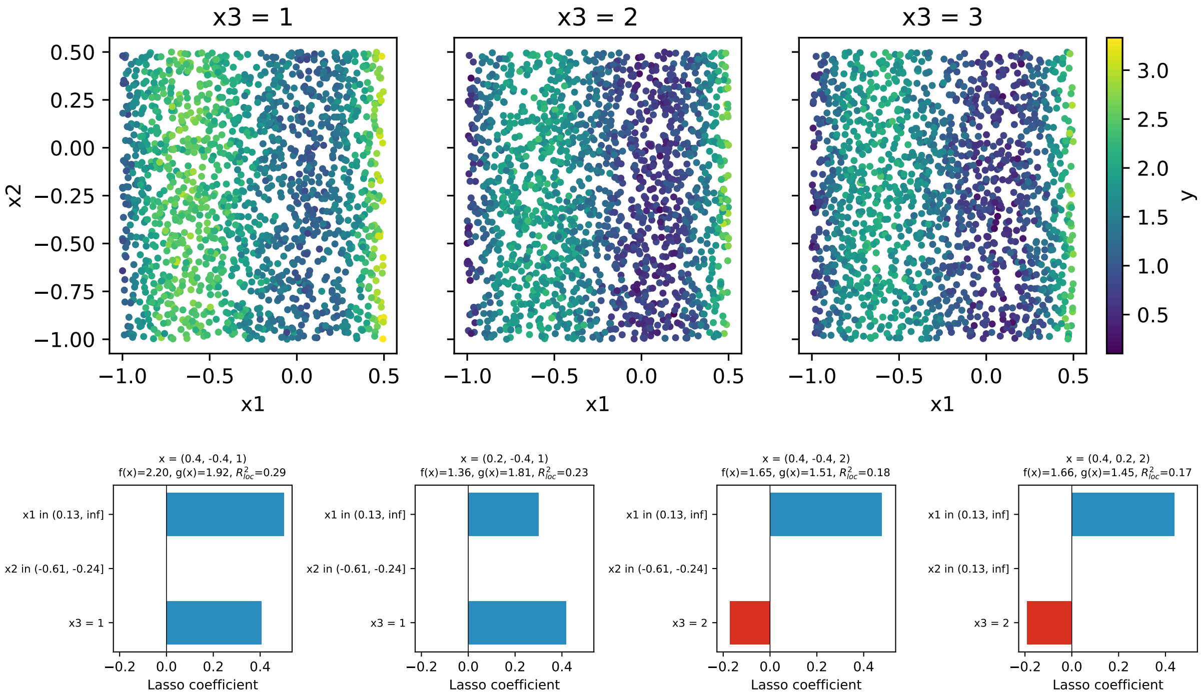

Figure 1: Top: y vs. (x_1, x_2) split by x_3, full data set. Bottom: LIME bar plots, blue = +, red = −. Each panel shows the black-box prediction $f(x)$, the surrogate prediction $g(x)$, and the kernel-weighted local R^2 of the surrogate.

2 Shapley values

The Shapley value of feature j at point x is

$$\varphi_j(x) \;=\; \sum_{S\subseteq N\setminus\{j\}} \frac{|S|!\,(d-|S|-1)!}{d!}\,\big(v(S\cup\{j\})-v(S)\big), \qquad v(S) \;=\; \mathbb{E}_{X\sim p}\big[f(X_S=x_S, X_{N\setminus S})\big],$$

the average marginal contribution of j over a uniformly random ordering of the features. Under the assumption that features are independent the coalition value $v(S)$ is the interventional expectation above, estimated by Monte Carlo over a background sample $B \subset X_{\text{train}}$: hold features in S fixed at x_S, replace the others with the rows of B, average the black-box prediction. We use $|B| = 500$ rows drawn uniformly from the training split; the variance of $\hat{v}(S)$ is already an order of magnitude below the inter-feature differences at this size.

With $d = 3$ the $2^d = 8$ coalition values can be enumerated exactly, so the only Monte-Carlo step is the background average. Two equivalent paths recover the same numbers and we implement both: the exact subset-weighted sum above, and a permutation-Monte-Carlo (Strumbelj and Kononenko, 2014) that draws random feature orderings and chains $b \to (b \text{ with } x \text{ inserted along } \pi) \to \cdots \to x$, accumulating one-step prediction differences. With 5000 permutations the latter matches the exact value to within 0.006 on every feature for every query point, which is a useful sanity check on the implementation.

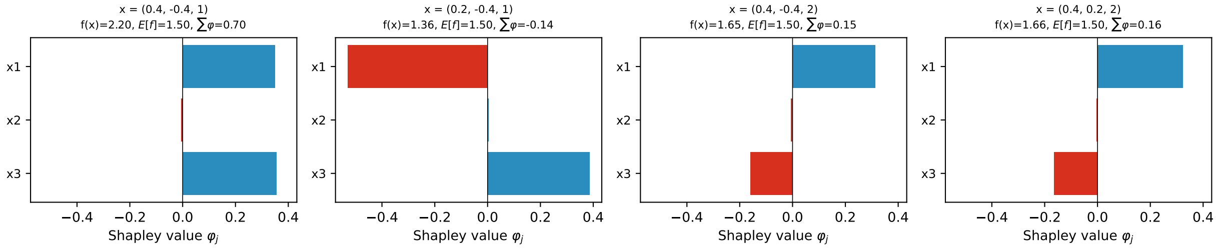

Figure 2: Shapley values $\varphi_j(x)$ for the four query points. Each title shows $f(x)$, the baseline $\mathbb{E}[f] = 1.50$, and $\sum_j \varphi_j$. By the efficiency axiom $f(x) - \mathbb{E}[f] = \sum_j \varphi_j$ exactly, and indeed every panel satisfies this to numerical precision.

Table 2: Exact Shapley values under feature independence, $\mathbb{E}[f] = 1.499$. The right-most column verifies the efficiency axiom $\sum_j \varphi_j = f(x) - \mathbb{E}[f]$ (to numerical precision).

x_1	x_2	x_3	$f(x)$	φ_{x_1}	φ_{x_2}	φ_{x_3}	$\sum_j \varphi_j$	$f - \mathbb{E}[f]$
0.4	−0.4	1	2.199	+0.349	−0.004	+0.355	+0.700	+0.700
0.2	−0.4	1	1.360	−0.530	+0.004	+0.387	−0.139	−0.139
0.4	−0.4	2	1.646	+0.313	−0.007	−0.159	+0.147	+0.147
0.4	0.2	2	1.656	+0.323	−0.003	−0.163	+0.157	+0.157

Shapley resolves what LIME with quartile bins cannot. The two instances $(0.4, -0.4, 1)$ and $(0.2, -0.4, 1)$ produce identical z vectors and so identical LIME explanations, but their Shapley attributions for x_1 differ in sign and magnitude ($+0.35$ vs. -0.53). The latter is consistent with the colour map in Fig. 1: at $x_1 = 0.2$ the model sits in the low-y trough between the two yellow bands, so x_1 contributes negatively relative to baseline; at $x_1 = 0.4$ the model is close enough to the right-hand yellow band that x_1 contributes positively. The x_3 attributions are $+0.36/+0.39/-0.16/-0.16$, the same sign pattern as the marginal $\bar{y} \mid x_3$ gap and within 0.05 of the LIME categorical effects. The x_2 attributions are below 0.01 in absolute value on all four instances, agreeing with the LIME finding that x_2 carries no signal locally.

The two methods answer different questions, which explains both their agreement on the x_3 effect and their divergence on samples 1 vs. 2. LIME fits a sparse linear surrogate to a kernel-weighted neighbourhood, so its coefficients describe a local linear trend in z space; Shapley distributes the gap $f(x) - \mathbb{E}[f]$ across features by averaging marginal contributions over all coalition orderings, with no surrogate model and no discretization step. On this dataset Shapley is closer to what one wants from a feature attribution: it is signed and continuous in x, it satisfies efficiency by construction, and it does not collapse two different x_1 values to the same bin.